

A positive semicontinuous density for the Syracuse stationary law

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Abstract

Let μ be the stationary probability measure on $\mathbb{Z}_3$ for $x \mapsto (3x+1)/2^a$, with independent geometric exponents $\mathbb{P}(a=j) = 2^{-j}$. We construct its greatest nonnegative lower semicontinuous density with respect to additive Haar probability. This density vanishes off the units, has a uniform positive lower bound on the units, and takes a prescribed arithmetic value at every positive integer. The value is the canonical trace obtained from weighted first-hit densities, including the cycle factor at periodic targets. Consequently every unit cylinder of depth k has mass at least $c3^{-k}$, with one constant $c > 0$, and the normalized minimum cylinder exponent in Tao's 2020 question equals one. At every positive integer the cylinder densities also converge to the arithmetic trace in Cesàro mean absolute error. The density is infinite on a dense Haar-null G_δ subset of the units, and its integer trace is unbounded in every unit residue class. The proof combines periodic first-hit means, a superharmonic bound on inverse layers, and a normalization argument for an arithmetic lower envelope. It does not settle forward Collatz convergence or the expected late-occupation bound.

1 The stationary measure and the arithmetic trace

Write λ for additive Haar probability on $\mathbb{Z}_3$, and $U = \mathbb{Z}_3^\times$, so $\lambda(U) = 2/3$. The Syracuse stationary measure is the law of

$$X = \sum_{j=0}^{\infty} 3^j 2^{-(a_1 + \dots + a_{j+1})}, \quad \mathbb{P}(a_i = a) = 2^{-a} \quad (a \geq 1),$$

where the exponents are independent. This is Tao's random variable [1]. Put

$$p_k(b) = \mu\{x : x \equiv b \pmod{3^k}\}, \quad \rho_k(x) = 3^k p_k(x \bmod 3^k).$$

All Haar densities below use λ , rather than Haar probability normalized on U .

For the arithmetic definitions use the shortcut map

$$T(n) = \begin{cases} n/2, & 2 \mid n, \\ (3n+1)/2, & 2 \nmid n. \end{cases}$$

Let $o_r(n)$ count odd entries among $n, Tn, \dots, T^{r-1}n$, and set

$$R_r(n) = 3^{o_r(n)} 2^{-r}.$$

If q reaches a positive target N , let $\tau_N(q)$ be its first hitting time and define

$$w_N(q) = \frac{q}{N} R_{\tau_N(q)}(q).$$

Set $w_N(q) = 0$ when q does not reach N . These weights lie in $[0, 1]$. Their natural means have limits, denoted by D_N . We use the arithmetic density and clock results in [4]; their upstream inputs include the separate constructions of Mazur and Said Duran described there.

If N has a positive return, let r_N be its first positive return time and put

$$G_N = (1 - R_{r_N}(N))^{-1}.$$

Otherwise put $G_N = 1$. Define

$$\delta = \frac{1}{2} \log \frac{4}{3}, \quad g(N) = \delta G_N N D_N. \quad (1)$$

Here a nonperiodic integer means an integer with no positive return to itself. It may enter a cycle later.

The construction in this paper starts from the actual function g , rather than a value assigned to an almost-everywhere density on the countable set of positive integers. For $x \in \mathbb{Z}_3$, set

$$e_k(x) = \inf_{\substack{n \in \mathbb{N}_{>0} \\ n \equiv x \pmod{3^k}}} g(n), \quad E(x) = \sup_{k \geq 0} e_k(x). \quad (2)$$

The functions e_k are finite and constant on depth- k cylinders. The function E takes values in $[0, \infty]$.

Theorem 1.1. *The function E has the following properties.*

1. *It is lower semicontinuous, $\int E d\lambda = 1$, and $\mu = E\lambda$.*
2. *It vanishes on $3\mathbb{Z}_3$. There is one constant $c > 0$ such that $E(x) \geq c$ for all $x \in U$.*
3. *At every positive integer, including every periodic integer, $E(n) = g(n)$.*
4. *If $h : \mathbb{Z}_3 \rightarrow [0, \infty]$ is lower semicontinuous and $h\lambda = \mu$, then $h \leq E$ everywhere.*

The greatest representative uses the extended nonnegative reals. Corollary 1.4 below shows that its infinite values cannot be removed while retaining the greatest-representative property.

Corollary 1.2. *There is a constant $c > 0$ such that*

$$p_k(b) \geq c3^{-k} \quad (k \geq 1, b \in \mathbb{Z}/3^k\mathbb{Z}, 3 \nmid b).$$

If $m_k = \min_{3 \nmid b} p_k(b)$, then

$$c3^{-k} \leq m_k \leq \frac{3}{2} 3^{-k}, \quad \lim_{k \rightarrow \infty} \frac{-\log m_k}{k \log 3} = 1.$$

Tao introduced this minimum and conjectured the exponent-one asymptotic in his 2020 discussion [2]. Nikpour and Rabbani's preprint [3] reports an upper bound of 1.039979 for this minimum-atom exponent. Corollary 1.2 gives a constant lower factor. The proof uses the arithmetic inputs behind (1); it is not a deduction from almost-everywhere integrability alone. We make no claim of an exhaustive priority search.

Corollary 1.3. *For every positive integer n ,*

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=0}^{K-1} |\rho_k(n) - g(n)| = 0. \quad (3)$$

Also $e_k(n) \uparrow g(n)$. In particular, for each $\varepsilon > 0$ there is a conductor j such that

$$m \in \mathbb{N}_{>0}, \quad m \equiv n \pmod{3^j} \implies g(m) > g(n) - \varepsilon.$$

The convergence in (3) concerns the integer trace of the cylinder densities. It does not assert convergence of the individual values $\rho_k(n)$. None of these conclusions excludes an additional cycle or a divergent positive orbit.

Corollary 1.4. *The set $\{x : E(x) = \infty\}$ is a dense G_δ subset of U and has Haar measure zero. In every residue class prime to three, g is unbounded on the positive odd integers in that class. The function E is discontinuous at every positive integer prime to three.*

2 First-hit means and a bound on inverse layers

We record the arithmetic inputs in the normalization used here. They are supplied by the native first-hit development and the periodic density extension accompanying [4].

The function g is finite and nonnegative on $\mathbb{N}_{>0}$, vanishes on multiples of three, and satisfies one uniform bound

$$g(n) \geq c_0 > 0 \quad (3 \nmid n). \quad (4)$$

Its binary harmonic identity is

$$g(n) = \frac{1}{2}g(2n) + \frac{3}{2}\mathbf{1}_{n \equiv 2 \pmod{3}}g((2n-1)/3). \quad (5)$$

The native canonical identification gives

$$\frac{1}{K} \sum_{k < K} \rho_k(n) \longrightarrow g(n) \quad (n \in \mathbb{N}_{>0}). \quad (6)$$

The cycle factor in (1) is necessary for both formulas at periodic targets.

For $K \geq 0$ put

$$C_{K,N}(q) = \mathbf{1}_{q \text{ odd, } q \text{ reaches } N} R_{\tau_N(q)}(q).$$

$o_{\tau_N(q)}(q) \leq K$

The indicator makes this zero when the first hitting time is undefined. For a function p of positive period M , write $\langle p \rangle = M^{-1} \sum_{q=0}^{M-1} p(q)$.

Proposition 2.1 (Periodic marked first-hit mean). *Let $N > 0$, and let $p : \mathbb{N} \rightarrow \mathbb{R}$ be bounded, nonnegative, periodic, and zero outside odd integers prime to three. Then*

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{q \geq 1} C_{K,N}(q)p(q) = 2\delta N D_N \langle p \rangle. \quad (7)$$

If N is nonperiodic, the right side is $2g(N)\langle p \rangle$.

Proof. The periodic source theorem gives the natural mean $D_N\langle p \rangle$ for $w_N(q)p(q)$, and partial summation gives the same logarithmic mean. For every $\varepsilon > 0$, the odd sources reaching N for which the bounds

$$(1 - \varepsilon) \frac{\log q}{2\delta} \leq o_{\tau_N(q)}(q) \leq (1 + \varepsilon) \frac{\log q}{2\delta}$$

fail form a set of absolute logarithmic density zero. For a bounded nonnegative mask, its exceptional contribution is bounded by a fixed multiple of the unmasked exceptional contribution. The same domination applies to the summable endpoint tail beyond the native exponential height cutoff. The two-sided clock squeeze therefore converts the logarithmic source mean to

$$\frac{1}{K} \sum_{q \geq 1} \mathbf{1}_{q \text{ odd, } q \text{ reaches } N} \frac{w_N(q)p(q)}{o_{\tau_N(q)}(q) \leq K} \longrightarrow 2\delta D_N\langle p \rangle.$$

Multiplication by N gives (7). The formal proof first treats masks bounded by one, then rescales a general bounded mask. It retains the summability and endpoint-tail hypotheses in the clock squeeze and discharges them by domination from the actual first-hit estimates. $\square$

Call a nonnegative function $f : \mathbb{N}_{>0} \rightarrow \mathbb{R}$ binary superharmonic if

$$\frac{1}{2}f(2n) + \frac{3}{2}\mathbf{1}_{n \equiv 2 \pmod{3}} f((2n-1)/3) \leq f(n) \quad (n > 0). \quad (8)$$

Lemma 2.2 (Inverse-layer bound). *If f is binary superharmonic and p is bounded, nonnegative, and satisfies $p(q) \leq f(q)$ for $q > 0$, then*

$$\sum_{q \geq 1} C_{K,N}(q)p(q) \leq (K+1)f(N) \quad (N > 0, K \geq 0). \quad (9)$$

Proof. First sum over all pairs (q, r) with q odd, $T^r q = N$, and $o_r(q) = k$, restricting r to a finite range. The weight of a pair is $R_r(q)p(q)$. For $k = 0$, the only possible contribution is the zero-time term, bounded by $f(N)$. For positive k , split according to the last edge into N . The even predecessor is $2N$, with factor $1/2$ and unchanged odd count. When it exists, the odd predecessor is $(2N-1)/3$, with factor $3/2$ and odd count reduced by one. Induction on the time cutoff, followed by (8), bounds each layer by $f(N)$. Nonnegative summation removes the time cutoff. Summing over $0 \leq k \leq K$ gives $(K+1)f(N)$. First hits form a subset of these pairs, proving the claim. The formal series rearrangements use the native summability of each unmarked odd-depth layer and boundedness of p . $\square$

Lemma 2.3 (Periodic test bound). *If p is bounded, nonnegative, periodic, zero outside odd units, and $p(q) \leq g(q)$ for positive q , then $\langle p \rangle \leq 1/2$.*

Proof. Apply Lemma 2.2 with $f = g$, which is harmonic. Use the positive nonperiodic target $N = 5$. Its shortcut orbit enters $1, 2$, so it never returns to five, and $g(5) > 0$ by (4). After division by K , Proposition 2.1 gives $2g(5)\langle p \rangle \leq g(5)$. Cancel $g(5)$. $\square$

3 The lower envelope has mass one

Every cylinder contains a positive integer, so e_k in (2) is finite and continuous. Cylinder refinement gives $e_k \leq e_{k+1}$, hence E is lower semicontinuous. Also $E(n) \leq g(n)$ for every positive integer. Every nonunit cylinder at conductor at least one contains only integer arguments where g vanishes. The unit lower bound already holds at conductor one. Therefore

$$E = 0 \text{ on } 3\mathbb{Z}_3, \quad E \geq c_0 \text{ on } U. \quad (10)$$

Lemma 3.1 (Maximality of the arithmetic envelope). *If $h : \mathbb{Z}_3 \rightarrow [0, \infty]$ is lower semicontinuous and $h(n) \leq g(n)$ at every positive integer, then $h \leq E$.*

Proof. If $a < h(x)$, lower semicontinuity supplies a cylinder about x on which $h > a$. Every positive integer in that cylinder has $g(n) > a$, so its infimum $e_k(x)$ is at least a . Take the supremum over $a < h(x)$. $\square$

For $k \geq 1$ the integer function $p(q) = \mathbf{1}_{q \text{ odd}} e_k(q)$ has period $2 \cdot 3^k$ and satisfies Lemma 2.3. Among each pair $q, q + 3^k$, exactly one integer is odd. Thus

$$\langle p \rangle = \frac{1}{2} \int e_k d\lambda \leq \frac{1}{2}.$$

The same integral bound at $k = 0$ follows by monotonicity. Consequently

$$0 < \beta := \int E d\lambda \leq 1. \quad (11)$$

Positivity uses (10); the upper bound uses monotone convergence.

Define the transfer operator on nonnegative functions by

$$(\mathcal{L}h)(x) = \frac{1}{2}h(2x) + \frac{3}{2}\mathbf{1}_{x \equiv 2 \pmod{3}}h((2x-1)/3).$$

The second inverse branch is used only on its clopen domain. This operator preserves lower semicontinuity and Haar integrals. Indeed, doubling preserves λ , while $y \mapsto (3y+1)/2$ maps $\mathbb{Z}_3$ onto the residue class two with Haar scaling $1/3$. At every positive integer n , the inequality $E \leq g$ and (5) give $(\mathcal{L}E)(n) \leq g(n)$. Lemma 3.1 therefore yields $\mathcal{L}E \leq E$ everywhere. Equal finite integrals imply $\mathcal{L}E = E$ almost everywhere.

It follows that $E\lambda$ is stationary for the binary maps $H(x) = x/2$ and $J(x) = (3x+1)/2$, each with probability $1/2$. Iterating the H part of this equation gives stationarity for $H^{a-1}J(x) = (3x+1)/2^a$, with weights 2^{-a} ; after a such terms the remaining mass is $\beta 2^{-a}$. Uniqueness of the stationary probability follows from the common 3-adic contraction factor of these affine maps. Hence

$$\beta^{-1}E\lambda = \mu. \quad (12)$$

Lemma 3.2 (Canonical lower semicontinuous minorants). *If h is a nonnegative lower semicontinuous density for μ , then $h(n) \leq g(n)$ for every positive integer n .*

Proof. For $a < h(n)$, all sufficiently fine cylinders about n have $h > a$. Their normalized masses therefore satisfy $\rho_k(n) \geq a$ for all sufficiently large k . Equation (6) gives $g(n) \geq a$. Let a increase to $h(n)$. $\square$

Apply this lemma to $h = E/\beta$, using (12). Maximality of the arithmetic envelope gives $E/\beta \leq E$. Since $0 < E(1) \leq g(1) < \infty$, evaluation at one implies $\beta \geq 1$. Together with (11), this proves

$$\beta = 1, \quad \mu = E\lambda.$$

Lemma 3.2 and Lemma 3.1 also show that every nonnegative lower semicontinuous canonical density lies below E .

4 Equality at the positive integers

The restriction $n \mapsto E(n)$ is finite, nonnegative, and binary superharmonic. Fix a positive non-periodic integer N . For $j \geq 1$, apply Lemma 2.2 with $f(n) = E(n)$ and $p(q) = \mathbf{1}_{q \text{ odd}} e_j(q)$. Proposition 2.1 and the exact odd-mask mean give

$$g(N) \int e_j d\lambda \leq E(N).$$

As $j \rightarrow \infty$, the integrals increase to one. Thus $g(N) \leq E(N)$. The reverse inequality is part of the construction, so $E(N) = g(N)$ at every positive nonperiodic integer, without a parity restriction.

To handle periodic points, put $d(n) = g(n) - E(n)$. This is nonnegative, is zero at nonperiodic points, and satisfies

$$d(n) \leq \frac{1}{2}d(2n) + \frac{3}{2}\mathbf{1}_{n \equiv 2 \pmod{3}}d((2n-1)/3).$$

A deterministic map has at most one periodic predecessor of any given point: iterate their common image for one less than a common multiple of their periods to recover either predecessor. Consequently, if n is periodic, all predecessors of Tn other than n have zero defect. It follows that

$$d(Tn) \leq R_1(n)d(n), \quad d(T^r n) \leq R_r(n)d(n).$$

For any positive return $T^r n = n$, the positive affine correction implies $R_r(n) < 1$. Thus $0 \leq d(n) \leq R_r(n)d(n)$ forces $d(n) = 0$. This completes Theorem 1.1.

5 Cylinder minima and trace consequences

A unit cylinder of depth $k \geq 1$ lies wholly in U . Integrating $E \geq c_0$ over it gives $p_k(b) \geq c_0 3^{-k}$. There are $2 \cdot 3^{k-1}$ unit cylinders and their masses sum to one. Their minimum is therefore at most $(2 \cdot 3^{k-1})^{-1}$. These bounds imply

$$-\log 3 + \frac{\log c_0}{k} \leq \frac{\log m_k}{k} \leq -\log 3 + \frac{\log(3/2)}{k},$$

proving Corollary 1.2.

At a positive integer n , lower semicontinuity and $E(n) = g(n)$ imply

$$\rho_k(n) \geq g(n) - \varepsilon \quad \text{for all sufficiently large } k.$$

Together with (6), this improves the signed Cesàro mean to the absolute-error mean. Explicitly, $\min(\rho_k(n), g(n)) \rightarrow g(n)$, and

$$|\rho_k(n) - g(n)| = \rho_k(n) + g(n) - 2\min(\rho_k(n), g(n)).$$

Average this identity to obtain (3). Finally, $e_k(n) \uparrow E(n) = g(n)$. Choosing one $e_j(n)$ above $g(n) - \varepsilon$, and using its definition as a cylinder infimum, proves the residue-class assertion in Corollary 1.3.

6 Dense infinite values

Write $H(x) = x/2$ and $J(x) = (3x+1)/2$ as before. The inequality $\mathcal{L}E \leq E$ gives, in the extended nonnegative reals,

$$E(Hx) \geq \frac{1}{2}E(x), \quad E(Jx) \geq \frac{3}{2}E(x).$$

Since $J(-1) = -1$ and $E(-1) > 0$, the second inequality forces $E(-1) = \infty$. The first then gives

$$E(-2^{-a}) = \infty \quad (a \geq 0).$$

These points are dense in U : powers of two represent every unit modulo every power of three. For completeness, elementary cubing gives $4^{3^r} = 1 + 3^{r+1}u_r$ with $u_r \equiv 1 \pmod{3}$. If a power of two represents a chosen residue modulo 3^k , multiplying it by one of $1, 4^{3^{k-1}}, 4^{2 \cdot 3^{k-1}}$ realizes each of the three lifts modulo 3^{k+1} . Start with the two units modulo three. Negation and inversion permute the unit residues, giving the claimed density of -2^{-a} .

Lower semicontinuity makes

$$\{E = \infty\} = \bigcap_{M \geq 1} \{E > M\}$$

a G_δ , and integrability makes it Haar null. Fix a unit cylinder and a finite threshold M . Choose an infinite-value point inside the cylinder. Lower semicontinuity supplies a smaller cylinder on which $E > M$; it contains a positive odd integer n , where $g(n) = E(n) > M$. This proves the unboundedness assertion. Finally, a finite value of E at a unit cannot be a continuity point because every neighborhood contains an infinite value. Theorem 1.1 supplies finiteness at each positive integer, proving Corollary 1.4.

Formalization and verification scope

The accompanying Lean development defines the same weighted first-hit density, cycle factor, arithmetic infima, and canonical stationary measure. The accompanying claim map gives the full theorem-to-source correspondence. Four principal endpoints in the namespace `CollatzCanonical.IntegerStationaryEnvelope` are:

```
canonicalIntegerEnvelope_withDensity
canonicalIntegerEnvelope_natCast_eq
canonicalMinimumUnitMass_base_three_rate
canonicalIntegerEnvelope_infinity_closure
```

The finite probability bound is stated directly for the native `syracPMF`, not just for an unrelated stationary model.

All 1,620 modules in the exact internal source closure and the generated declaration audit were freshly compiled with warnings treated as errors and individually checked by Lean’s official kernel checker: 3,242 successful commands. The audit records 36,651 module-header/constant rows, 36,539 distinct names, and 3,419 encoded-private rows. The only permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. Complete inventories of 53,006 external compiled objects agree before and after. Those libraries were fingerprinted and reused; neither an external rebuild nor correspondence with adjacent source checkouts is claimed. Setup reuse, fresh-process record verification, and verification after relocation passed. Relocation copied the internal objects and reused the same external libraries; it did not recompile the sources. The fresh dependency-download branch was not exercised.

Attribution and use of AI

The stationary Syracuse law, the first-passage framework, and the minimum-cylinder question are due to Tao. The arithmetic inputs are those of the companion basin-density development, with its

separate attribution to Mazur and Said Duran. The contribution here is the arithmetic-envelope construction, its normalization and pointwise identification, and the stated consequences.

OpenAI Codex was used extensively for mathematical exploration, Lean development, automated review, and preparation of this manuscript. Automated checking and review are not external peer review or endorsement by the authors of the cited works.

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